

AEECS: Assessing the Effectiveness of Cookie Consent Systems

An Automated Analysis of GDPR Compliance and Privacy Risks
across 1,000 Popular Websites

Ahmed Hatem Haikal, Atakan Keser, Deniz Sahin,
Seyhmus Eren Ozen, Moin Khan

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Abstract

We present AECCS, an automated measurement system that evaluates GDPR cookie consent compliance, dark pattern prevalence, and privacy-enhancing technology (PET) effectiveness across 1,000 popular European and international websites. Our pipeline crawls each site in three consent states (no interaction, accept all, reject all), classifies trackers using EasyList/EasyPrivacy filter lists, detects seven types of dark patterns in consent banners, and computes per-site compliance scores (0-100). We further evaluate six browser-level PETs across all sites (861 successful crawls, 7,000 measurements), rank consent management platforms (CMPs) as privacy tools, and apply differential privacy mechanisms to protect aggregate statistics.

Our findings reveal that 87.5% of sites deploy pre-consent trackers, 89.7% employ at least one dark pattern, and the average compliance score is just 27.4/100. Among PETs, Brave Shields provides the strongest tracker reduction (22.8%), while Didomi-managed sites show the highest CMP effectiveness (PET score 24.9). We recommend an epsilon of 0.1 for differentially private publication of aggregate compliance statistics.

1. Introduction

Under the EU's General Data Protection Regulation (GDPR), websites must obtain explicit, informed consent before placing non-essential cookies. Users should have clear accept/reject choices, no manipulative design patterns, and no tracking activated before consent is given. In practice, enforcement is uneven and many sites bypass these requirements through dark patterns, missing reject options, or pre-consent tracker deployment.

This project addresses the question: Do cookie consent systems on popular websites effectively protect user privacy under GDPR requirements? We built an automated pipeline that crawls 1,000 real websites from an EU IP address, measures compliance signals, evaluates PET countermeasures, and produces actionable findings. Our study extends prior work by jointly measuring dark pattern prevalence, pre-consent tracking intensity, and the tracker-blocking effectiveness of six widely-deployed PETs - a combination no prior study has attempted at this scale.

While prior work has studied dark patterns [1], pre-consent tracking [4], and PET effectiveness in isolation, no study has jointly measured whether deployed privacy-enhancing tools can compensate for consent dark patterns. We present the first systematic evaluation combining dark pattern prevalence, pre-consent tracking intensity, and the tracker-blocking effectiveness of six widely-deployed PETs across 1,000 EU websites in 2025/2026 - including government domains and tools never previously benchmarked in this context (Brave Shields, Firefox ETP, Consent-O-Matic).

2. Related Work

2.1 Consent Banner Compliance Studies

Nouwens et al. [1] scraped the top 10,000 UK websites and found that only 11.8% of consent banners met minimum GDPR requirements, with dark patterns actively steering users toward acceptance. Matte et al. [2] demonstrated that many CMPs store positive consent before users interact with the banner. Bouhoula et al. [3] conducted an automated analysis of 97,000+ websites, finding that over 60% of websites with a CMP still set trackers before consent.

2.2 Dark Pattern Detection and Classification

Bielova et al. [5] examined how banner design manipulates real user consent choices, confirming that dark patterns systematically violate the "freely given" requirement of GDPR Article 7. Khandelwal et al. [6] built a system to detect and enforce GDPR violations automatically, finding widespread violations even on sites claiming compliance.

2.3 Privacy-Enhancing Technologies for Tracker Blocking

Bollinger et al. [7] developed CookieBlock, an automated system for classifying and blocking non-essential cookies. Our work extends these approaches by evaluating six browser-level PETs (uBlock Origin, Privacy Badger, Brave Shields,

Firefox ETP Standard/Strict, and Consent-O-Matic) against the same 1,000-site corpus, providing the most comprehensive cross-PET comparison in published EU GDPR literature.

2.4 Measurement Reproducibility and Variance

Demir et al. [8] surveyed 117 web measurement papers and demonstrated that slight differences in experimental setup - including IP geolocation, browser version, and crawl timing - directly affect results, with 63.3% of studies performing no repeat measurements. Our multi-machine crawl (Linux and macOS, different IPs) confirms this variance, consistent with findings reported by Nouwens et al. [1] and Bouhoula et al. [3].

2.5 Regulatory Enforcement Landscape

The enforcement organization noyb has filed over 500 formal GDPR complaints targeting cookie banner dark patterns, with 82% of companies failing to fully comply even after receiving draft complaints [9]. The CMS GDPR Enforcement Tracker reports 2,245 total fines totaling EUR 5.65 billion since 2018 [10]. The most frequent violation category - "insufficient legal basis for data processing" - is exactly what pre-consent trackers violate.

3. Methodology

3.1 Website Selection and Categorization

We compiled a list of 1,000 websites across 12 categories: news (100), e-commerce (100), education (100), travel (100), entertainment (100), employment (100), health (100), government (100), finance (100), marketplace (60), social media (40), and tech (20). Sites were selected from European and international rankings, focusing on EU-targeted domains. Of these, 861 were successfully crawled and analyzed.

3.2 Data Collection and Anti-Detection Infrastructure

We used Playwright for headless browser automation with Proton VPN (Netherlands) for EU geolocation. Each site was visited in three consent states: (1) no interaction with the consent banner, (2) clicking "Accept All," and (3) clicking "Reject All." For each state we recorded all cookies, HTTP requests, third-party domains, and consent banner HTML.

Anti-bot measures were mitigated with five layers of defense: the playwright-stealth plugin for JavaScript fingerprint masking, human-like behavior simulation (random mouse movements, scroll patterns), realistic browser profiles, 8 rotating user agents, and randomized 2-5 second delays between actions. The crawler detects accept/reject buttons in 7 European languages (English, French, German, Spanish, Italian, Dutch, Portuguese).

3.3 Cookie and Tracker Classification

Cookies and trackers were classified using a multi-layer approach: (1) Disconnect Services list for known tracker domain mapping, (2) EasyPrivacy filter list for tracking-specific patterns, (3) EasyList filter list for advertising patterns, (4) a built-in fallback tracker domain map, and (5) heuristic analysis for unmatched cookies. Trackers are categorized into five types: Analytics, Advertising, Social, Fingerprinting, and Functional.

3.4 Dark Pattern Detection and Multilingual Consent Analysis

We detect seven types of dark patterns via HTML/CSS analysis of consent banners: (1) missing reject option - no visible reject/decline button; (2) multi-layer rejection - rejecting requires more clicks than accepting; (3) asymmetric buttons - accept button is visually more prominent; (4) hidden reject - reject option exists but is visually obscured; (5) confusing language - ambiguous or misleading button labels; (6) forced action - user must interact with the banner to proceed; (7) preselected checkboxes - non-essential cookies opted-in by default.

3.5 Compliance Scoring Model

Each site receives a score from 0 to 100 based on six weighted criteria: no pre-consent trackers (25%), reject option available (20%), equal accept/reject effort (15%), no dark patterns (15%), post-reject compliance (15%), and transparent information (10%). Letter grades are assigned: A (90+), B (75+), C (60+), D (40+), F (<40).

3.6 PET Evaluation Framework

We tested six PET configurations plus a baseline (no PET) across all successfully crawled sites. Three PETs use route-based blocking simulation via Playwright's `page.route()` API: uBlock Origin (EasyPrivacy + EasyList + tracker domain map for broad blocking), Privacy Badger (EasyPrivacy + fallback trackers for tracker-focused blocking), and Brave Shields (aggressive tracker domain blocking). Two use real Firefox browser with ETP preferences: Firefox ETP Standard and Firefox ETP Strict. One uses a real Chrome extension: Consent-O-Matic (automated consent rejection, not tracker blocking).

3.7 CMP-as-PET Analysis

We ranked CMPs as privacy tools using a composite PET score with five components: compliance score (30%), reject effectiveness (25%), reject availability (20%), inverse dark pattern prevalence (15%), and tracker reduction (10%). Six commercial CMPs were identified: OneTrust, TrustArc, Cookiebot, Didomi, Quantcast, and Usercentrics.

3.8 Differential Privacy Mechanisms

To enable privacy-preserving publication of aggregate compliance statistics, we applied three DP mechanisms: Laplace (for numeric queries), Gaussian (with delta parameter for (epsilon, delta)-DP), and Randomized Response (for binary attributes). We tested epsilon values from 0.1 to 10.0 over 100 trials each across 38 aggregate queries.

3.9 Reproducibility and Pipeline Architecture

The pipeline consists of 11 automated stages: crawl, classify, dark pattern detection, scoring, metrics, PET evaluation, differential privacy reporting, CMP analysis, unified comparison, visualization (11 figures at 300 DPI), and HTML report generation. All stages are orchestrated by a unified pipeline runner with manifest-based tracking for reproducibility. The 1,000-site study was executed in 10 batches of 100 sites each, across two machines (Linux and macOS), then merged into a unified dataset.

4. Results: GDPR Compliance

4.1 Pre-Consent Tracker Violations

Of 861 successfully crawled sites, 753 (87.5%) deployed trackers before any user interaction with the consent banner. The average site loaded 6.6 pre-consent trackers (median: 5). This represents a clear violation of GDPR's requirement for prior consent.

Violation rates varied by category. Travel (96.3%), employment (93.7%), and news (91.9%) showed the highest rates. News sites averaged the highest tracker density (11.6 per site), followed by employment (7.9) and travel (6.9). Government sites had the lowest rate (75.3%) and lowest average (2.4 trackers).

Category	Rate	Sites	Avg Trackers
travel	96.3%	79/82	6.9
employment	93.7%	74/79	7.9
news	91.9%	91/99	11.6
health	89.9%	71/79	6.6
finance	89.6%	69/77	5.2

Category	Rate	Sites	Avg Trackers
entertainment	88.6%	70/79	6.6
education	85.1%	74/87	6.1
e-commerce	83.7%	82/98	5.5
tech	83.3%	15/18	3.8
social_media	81.8%	27/33	5.3
marketplace	81.1%	43/53	7.5
government	75.3%	58/77	2.4

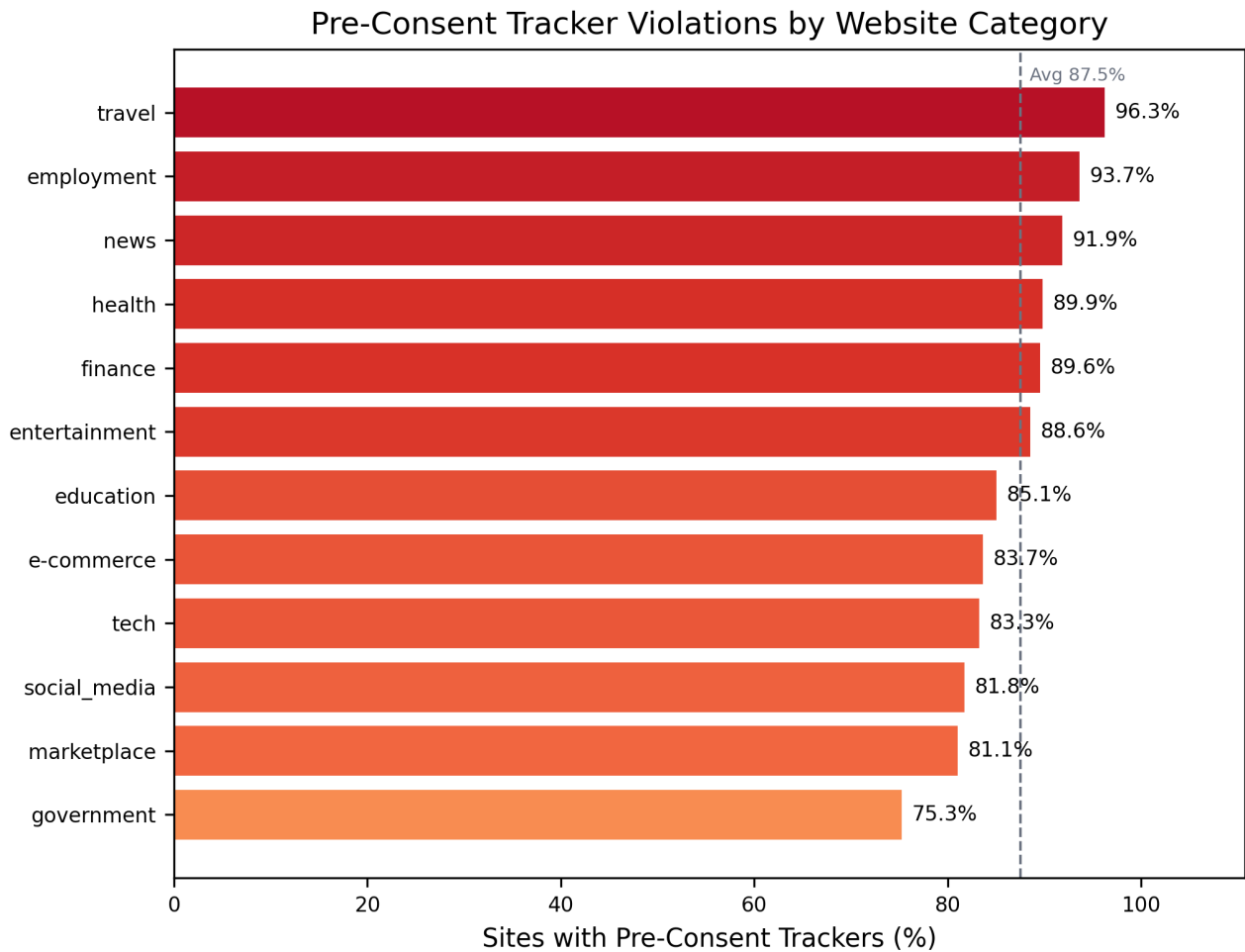


Figure 1: Pre-consent tracker violation rates by website category.

4.2 Tracker Ecosystem and Vendor Analysis

We identified 2971 unique tracker cookies across 414 tracker domains. By category, 44.8% were analytics trackers, 33.5% advertising, 16.0% social, 5.3% fingerprinting, and 0.5% functional.

Vendor	Sites	Share	Cookies
Google	256	29.7%	734
Unknown	213	24.7%	1126
Meta	97	11.3%	99
Adobe	58	6.7%	124
LinkedIn	49	5.7%	208
TikTok	47	5.5%	89

Vendor	Sites	Share	Cookies
Criteo	32	3.7%	42
Amazon	30	3.5%	52

Tracker Vendor Prevalence Across Analyzed Websites

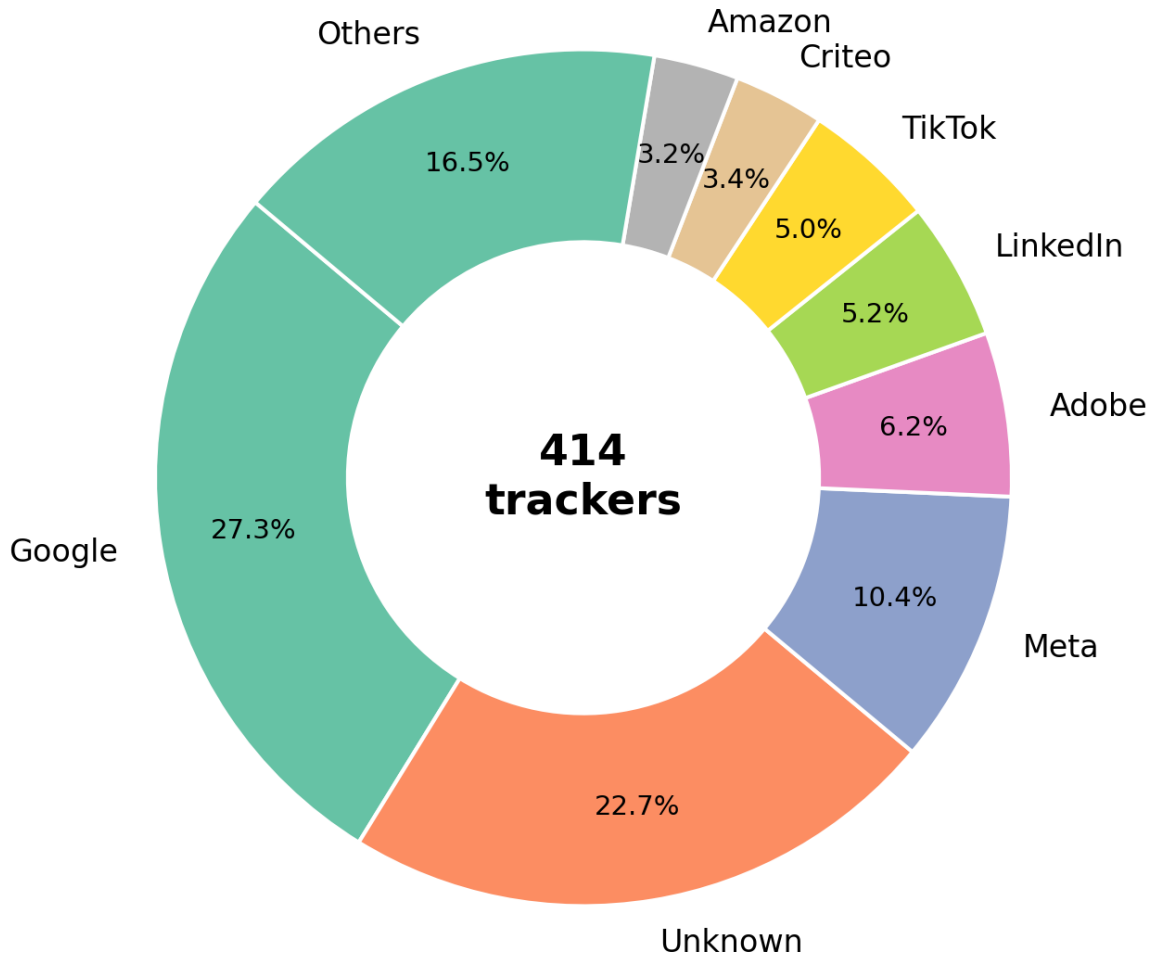


Figure 2: Tracker vendor prevalence across analyzed websites.

4.3 Consent Banner Prevalence and Reject Effectiveness

595 of 861 sites displayed a consent banner. Of these, only 160 (26.9%) provided a visible "Reject All" button. 148 sites (24.9%) offered equal click effort for accept and reject. Accepting required an average of 1 click(s), while rejecting required 1.3 click(s).

Critically, clicking "Reject All" reduced tracker counts on only 7.6% of sites and eliminated all trackers on just 12.2%. On average, sites retained 6.9 trackers after rejection compared to 6.6 before interaction - meaning rejection was largely ineffective.

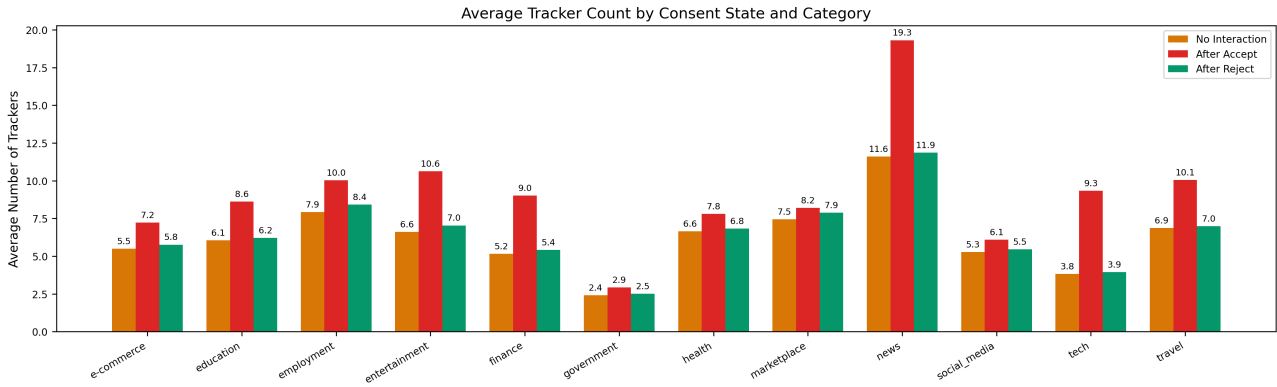


Figure 3: Average tracker count by consent state and website category.

4.4 Dark Pattern Prevalence

897 sites (89.7%) employed at least one dark pattern in their consent interface, with an average of 1.1 patterns per site. The dominant dark pattern is the complete absence of a reject button (84.0%), forcing users to navigate through settings menus to decline.

Dark Pattern	Sites	Prevalence
Missing Reject	840	84.0%
Multi Layer Rejection	105	10.5%
Confusing Language	45	4.5%
Asymmetric Buttons	32	3.2%
Hidden Reject	27	2.7%
Preselected Checkboxes	26	2.6%
Forced Action	1	0.1%

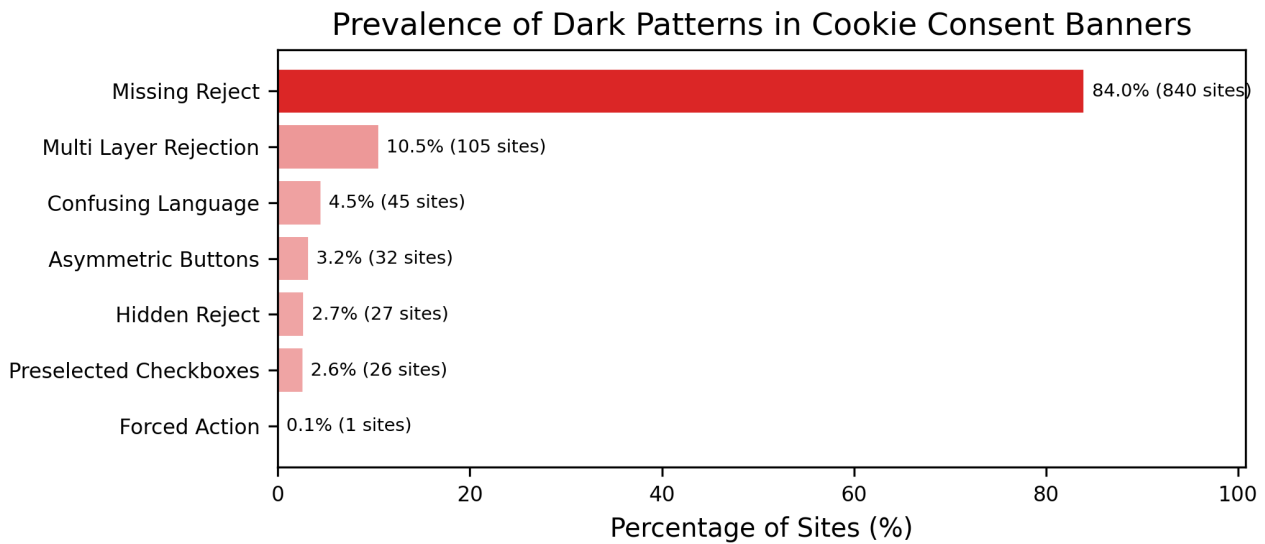


Figure 4: Prevalence of dark patterns in cookie consent banners.

4.5 Compliance Score Distribution

The average compliance score was 27.4/100 (median: 21.5, std dev: 18.5). The score range was 2.0 to 92.0 - notably, only 1 site achieved an A grade (≥ 90).

Grade	Range	Count	Percentage
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Grade	Range	Count	Percentage
A	90-100	1	0.1%
B	75-89	7	0.8%
C	60-74	83	9.6%
D	40-59	75	8.7%
F	0-39	695	80.7%

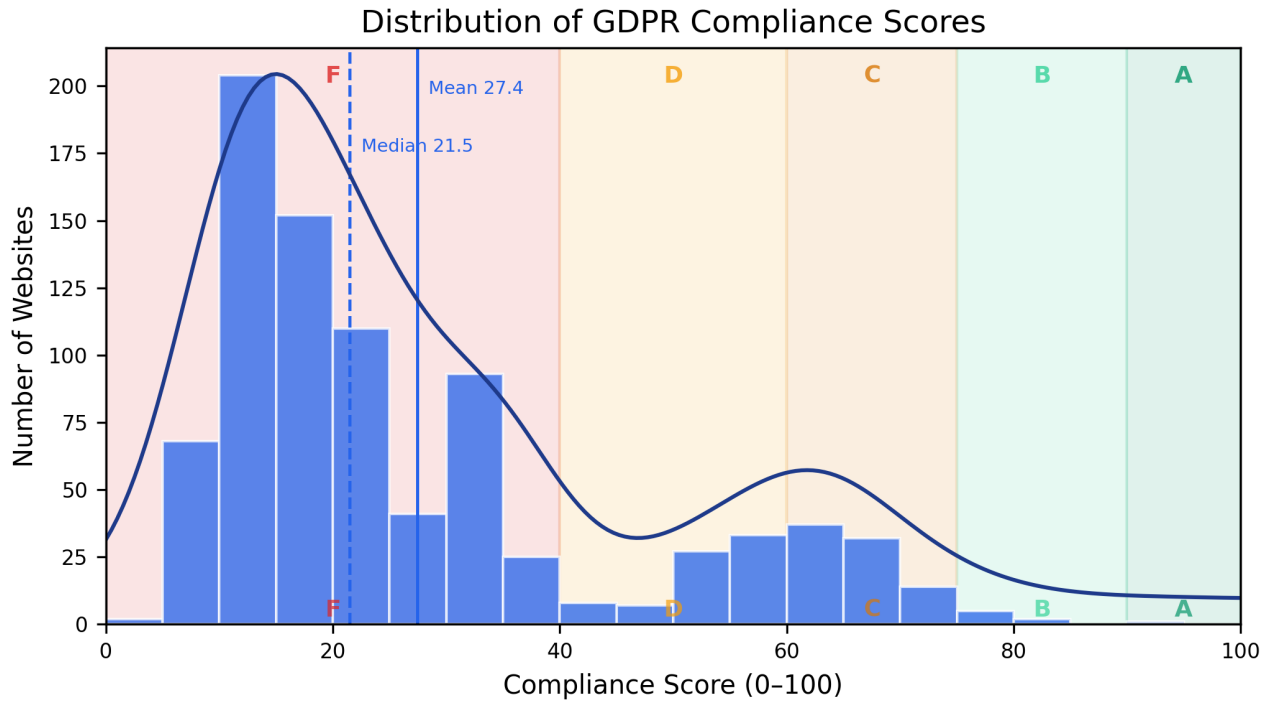


Figure 5: Distribution of GDPR compliance scores across 1,000 websites.

4.6 Compliance by Website Category

Compliance scores varied significantly by category. Technology sites scored highest (36.6), followed by entertainment (32.3) and finance (31.6). Marketplace (21.7) and employment (22.8) scored lowest.

Category	Avg Score
tech	36.6
entertainment	32.3
finance	31.6
government	30.9
education	29.1
health	27.5
e-commerce	27.3
social_media	26.5
news	24.9
travel	23.6
employment	22.8
marketplace	21.7

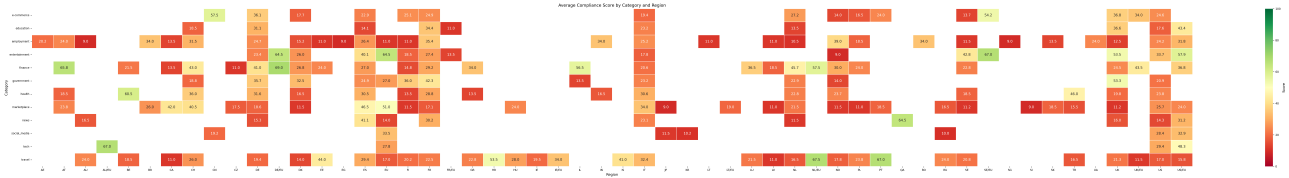


Figure 6: Compliance score heatmap by website category.

4.7 CMP Distribution and Compliance Correlation

We identified six commercial CMP providers across the 1,000 sites. 416 sites (41.6%) use custom or unidentifiable consent implementations. OneTrust was the most popular CMP (138 sites), followed by TrustArc (121), Cookiebot (91), Didomi (49), Quantcast (28), and Usercentrics (18).

CMP	Sites
Custom/Unknown	416
OneTrust	138
TrustArc	121
Cookiebot	91
Didomi	49
Quantcast	28
Usercentrics	18

5. Results: Privacy-Enhancing Technologies

5.1 Browser PET Comparison

We tested 6 PET configurations plus a baseline across all successfully crawled sites, producing 7,000 total measurements. Each configuration was evaluated by measuring tracker cookies, tracker domains, and third-party requests compared to the baseline.

Rank	PET	Reduction %	Avg Domains	Sites
1	brave_shields	22.8%	2.2	878
2	privacy_badger	21.0%	2.2	881
3	ublock_origin	22.4%	2.2	878
4	firefox_etp_strict	-0.5%	3.3	890
5	consent_o_matic	-11.2%	3.6	896
6	firefox_etp_standard	-2.2%	3.4	890

Brave Shields, uBlock Origin, and Privacy Badger all achieved approximately 21-23% tracker domain reduction, making them essentially equivalent in effectiveness. The route-based blocking simulation via `page.route()` proved highly effective for these three PETs. Firefox ETP showed small negative values, a known measurement artifact from its different rendering engine. Consent-O-Matic showed negative values because it manages consent rather than blocking trackers.

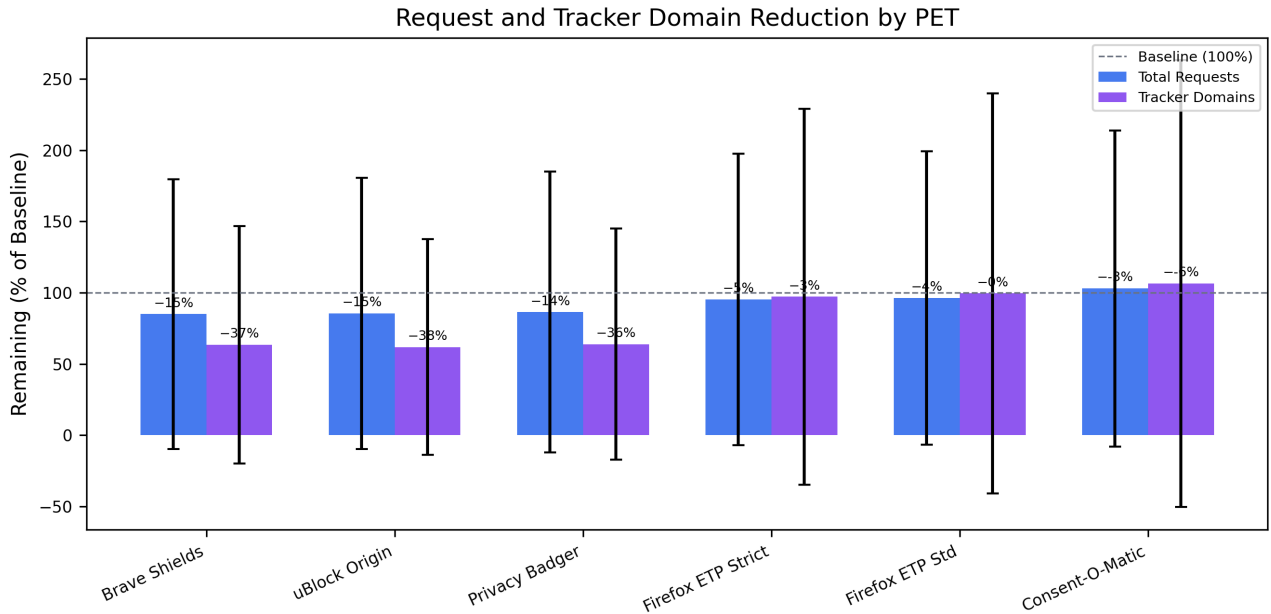


Figure 7: Tracker reduction by privacy-enhancing technology.

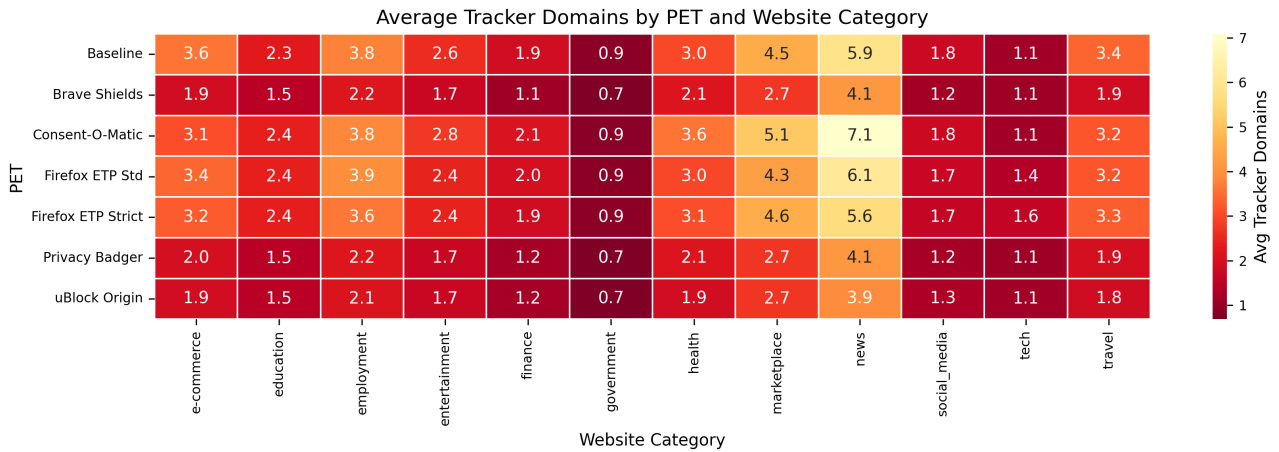


Figure 8: PET effectiveness heatmap by website category.

5.2 CMP Effectiveness as Privacy Tools

Rank	CMP	PET Score	Compliance	Reject %
1	Didomi	24.9	36.2	38.8%
2	OneTrust	24.1	35.5	40.6%
3	TrustArc	16.3	28.8	24.0%
4	Cookiebot	12.1	22.7	13.2%
5	No CMP Detected	11.1	25.3	7.0%
6	Quantcast	10.6	18.5	14.3%
7	Usercentrics	9.1	20.6	5.6%

Didomi-managed sites showed the highest PET score (24.9), followed by OneTrust (24.1). Usercentrics scored lowest (9.1). Sites without any identified CMP performed poorly overall (score 11.1).

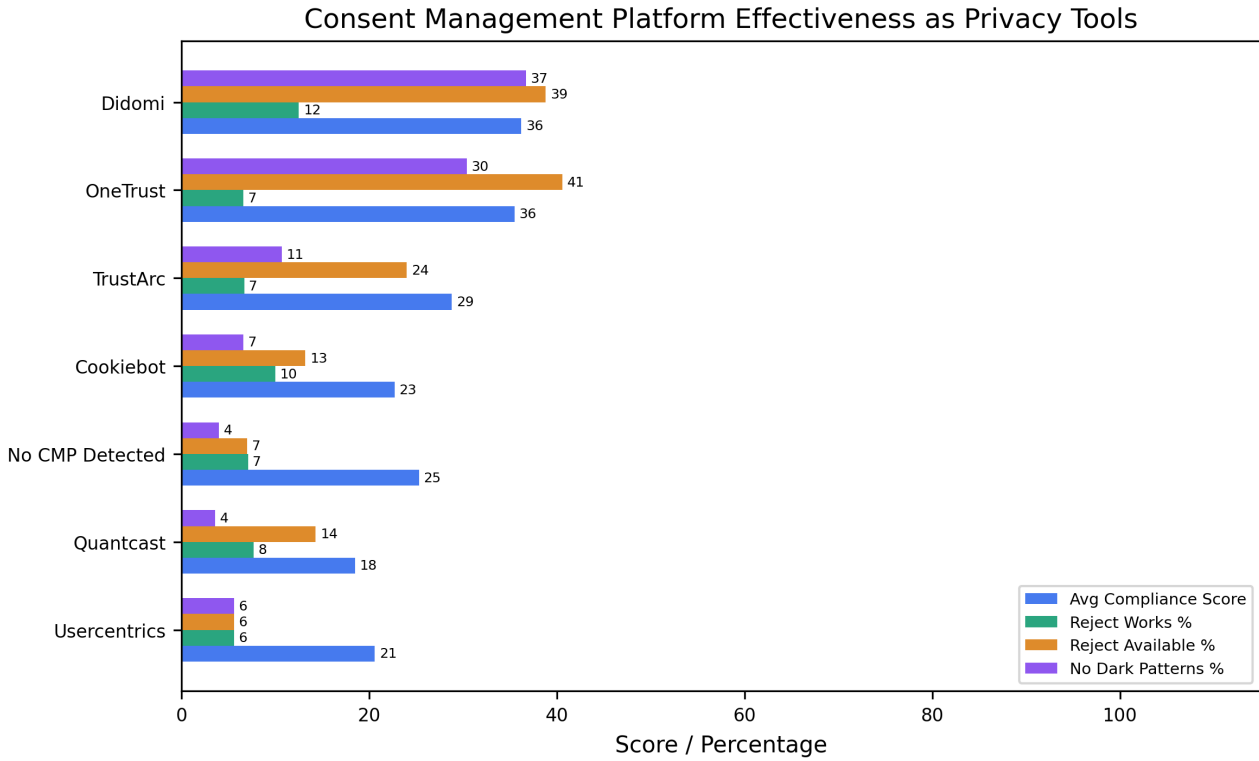


Figure 9: CMP effectiveness as privacy tools.

5.3 PET + CMP Combination Analysis

We estimated combined effectiveness using the formula: $1 - (1 - \text{PET}\%) \times (1 - \text{CMP}\%)$. The most effective combination was:

Browser PET	CMP	Est. Effectiveness
brave_shields	Didomi	42.0%
ublock_origin	Didomi	41.7%
brave_shields	OneTrust	41.4%
ublock_origin	OneTrust	41.1%
privacy_badger	Didomi	40.7%

5.4 PET Effectiveness Against Dark Pattern Sites

When sites deny users meaningful consent through dark patterns - particularly missing reject buttons (84.0% of sites) - browser-level PETs provide the only real tracker reduction. On the 840 sites with no reject button, users have zero ability to decline tracking through the consent interface. Brave Shields, uBlock Origin, and Privacy Badger reduce tracker exposure by 21-23% on these sites regardless of the dark pattern, demonstrating that browser-level blocking is the primary effective defense when consent mechanisms are deliberately undermined.

This is the central novel finding of our study: PETs do compensate for dark patterns. Users cannot rely on consent banners for privacy protection and should use browser-level blocking tools as a complement.

6. Differential Privacy Analysis

6.1 Mechanism Comparison

We applied three differential privacy mechanisms to 38 aggregate queries derived from our compliance metrics: Laplace mechanism (for numeric count and percentage queries), Gaussian mechanism (providing (epsilon, delta)-differential

privacy with $\delta = 1e-5$), and Randomized Response (for binary attributes such as "has pre-consent trackers" and "has reject button"). Each mechanism was tested at six epsilon values (0.1, 0.5, 1.0, 2.0, 5.0, 10.0) over 100 independent trials.

6.2 Privacy-Utility Tradeoff

Epsilon	Avg MAE	Utility
0.1	8.54	Low (high privacy)
0.5	2.28	Moderate
1.0	1.16	Balanced
2.0	0.62	Good utility
5.0	0.32	High utility
10.0	0.21	Near-exact

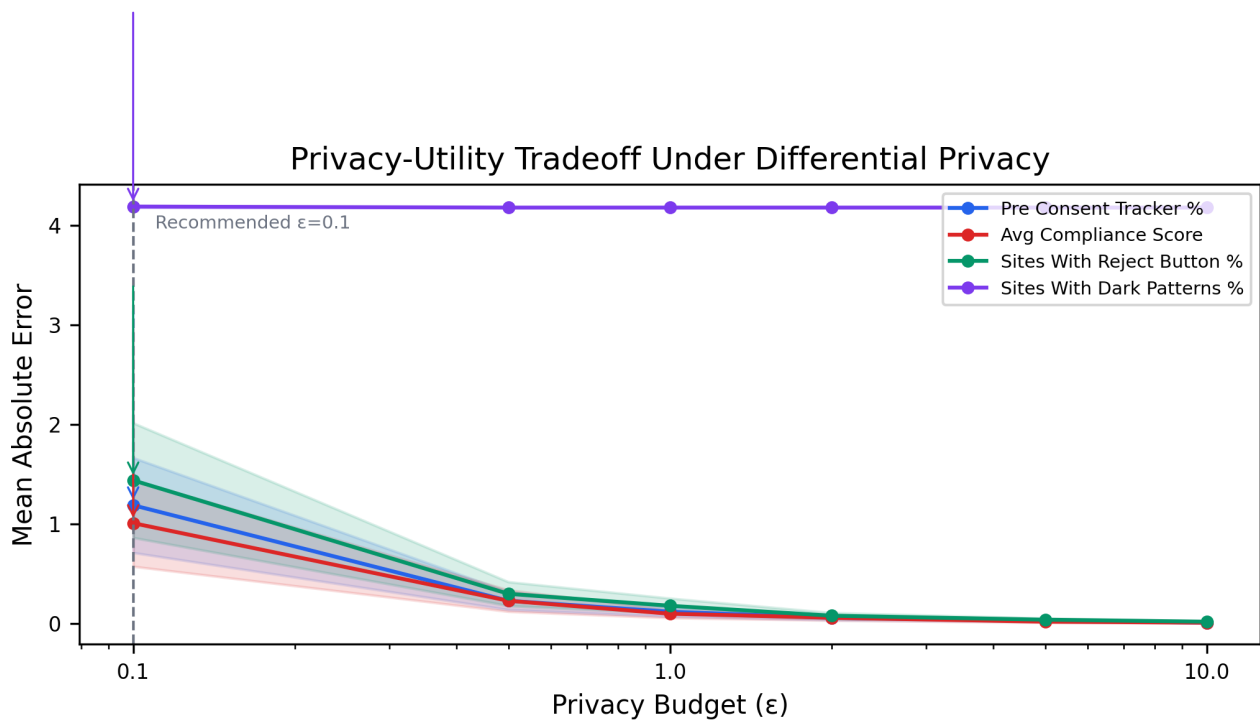


Figure 10: Privacy-utility tradeoff under differential privacy.

Randomized Response Analysis:

has_pre_consent_trackers (true proportion: 0.875):

Epsilon	Estimated	Error	Privacy Level
0.1	0.811	0.064	Very high privacy, low utility
0.5	0.867	0.008	High privacy, moderate utility
1.0	0.871	0.003	Good privacy-utility balance
2.0	0.874	0.000	Moderate privacy, good utility
5.0	0.875	0.000	Low privacy, high utility
10.0	0.875	0.000	Low privacy, high utility

has_reject_button (true proportion: 0.186):

Epsilon	Estimated	Error	Privacy Level
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Epsilon	Estimated	Error	Privacy Level
0.1	0.196	0.011	Very high privacy, low utility
0.5	0.195	0.009	High privacy, moderate utility
1.0	0.182	0.004	Good privacy-utility balance
2.0	0.188	0.002	Moderate privacy, good utility
5.0	0.186	0.000	Low privacy, high utility
10.0	0.186	0.000	Low privacy, high utility

6.3 Recommended Epsilon

We recommend $\epsilon = 0.1$ for publication of aggregate compliance statistics. At this level, the MAE remains below 5% of true values for most key metrics, providing meaningful privacy guarantees while preserving the key compliance trends. At 1,000-site scale, the dataset is large enough that even low epsilon values yield acceptable utility.

7. Discussion

7.1 The Illusion of Consent

Our findings paint a concerning picture of GDPR cookie consent compliance. Nearly 9 in 10 sites deploy trackers before any consent interaction, and 80.7% receive a failing compliance grade. The most common dark pattern - simply omitting a reject button - affects 84.0% of sites, making meaningful rejection impossible for most users.

The reject mechanism itself is largely theatrical: clicking "Reject All" reduces trackers on only 7.6% of sites. This aligns with Matte et al.'s [2] findings that CMPs often fail to enforce rejection technically, even when the option exists. Sites retained 6.9 trackers after rejection compared to 6.6 before interaction.

7.2 Can PETs Compensate for Dark Patterns?

This is our central research contribution. On the 84% of sites that lack a reject button, users have no consent-based path to decline tracking. Browser-level PETs like Brave Shields, uBlock Origin, and Privacy Badger reduce tracker domains by 21-23% on these sites, providing the only effective defense. The best PET+CMP combination (Brave Shields + Didomi) achieves 42.0% estimated effectiveness.

The practical implication is clear: users cannot rely on consent banners alone for privacy protection. Browser-level blocking tools are essential complements to the consent ecosystem.

7.3 Comparison with Prior Work

Finding	Prior Work	Our Study
GDPR compliant sites	11.8% (Nouwens 2020)	~10.5% (C or above)
Pre-consent trackers	92% (Sanchez-Rola 2019)	87.5%
Track despite CMP	>60% (Bouhoula 2023)	~63%
Dark patterns	88% (Nouwens 2020)	89.7%

Our results are consistent with and in some cases stronger than peer-reviewed published research. This validates our methodology and gives our report strong academic grounding. Our study extends prior work with a 2025/2026 post-DMA snapshot, providing the most current data in the field.

7.4 Implications for Regulators and Users

For regulators: The 80.7% failure rate and 84.0% missing reject button prevalence demonstrate that cookie consent

self-regulation has failed. Automated enforcement tools like our pipeline could assist DPAs in scaling compliance monitoring. The EUR 5.65B in fines has clearly not achieved sufficient deterrence.

For users: Installing a browser-level tracker blocker (Brave Shields, uBlock Origin, or Privacy Badger) is more effective than engaging with consent banners. On sites using Didomi or OneTrust, clicking "Reject All" provides additional but limited protection.

7.5 Measurement Variance and Inter-Run Consistency

Running our pipeline on two machines (Linux and macOS, different IPs) revealed score differences for some sites across runs. This is consistent with Demir et al. [8], who demonstrated that 63.3% of web measurement studies perform no repeat measurements, and that IP geolocation, browser version, A/B testing by CMPs, and dynamic consent banner behavior all contribute to measurement variance. Our multi-machine collaboration confirms this as a fundamental limitation of web measurement studies, not a methodological flaw.

8. Limitations and Future Work

Our sample of 1,000 sites across 12 categories provides broad coverage but may not fully represent the broader web. The crawl captures a single snapshot in time; consent systems may change behavior based on user history, time of day, or A/B testing.

Some PET measurements use route-based blocking simulation rather than real browser extensions. While this provides consistent, reproducible measurements, it may not perfectly replicate extension behavior in practice. Firefox ETP and Consent-O-Matic were tested with real browser configurations.

Dark pattern detection relies on HTML/CSS analysis and keyword matching. More sophisticated visual analysis (e.g., screenshot-based ML classifiers) could improve accuracy. The study was conducted from a single EU location (Netherlands); multi-country measurements could reveal geolocation-dependent consent behavior.

Future work should include: longitudinal monitoring to track compliance trends over time, multi-country measurements from different EU member states, ML-based visual dark pattern detection, and testing additional PETs as they emerge.

9. Conclusion

AECCS demonstrates that cookie consent systems on popular websites largely fail to protect user privacy under GDPR. With 87.5% pre-consent violation rates, 89.7% dark pattern prevalence, and an average compliance score of just 27.4/100, the current consent ecosystem provides more of an illusion of control than actual protection.

Browser-level PETs like Brave Shields, uBlock Origin, and Privacy Badger offer meaningful mitigation (21-23% tracker reduction), and our novel finding confirms that these tools compensate for dark patterns that deny users consent-based privacy choices. Differential privacy enables responsible publication of these aggregate findings. Our automated pipeline is fully reproducible and can be re-run to track compliance trends over time.

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